

PART A

Question 1:

BEGIN

INPUT N

FOR count $\leftarrow$ N DOWNTO 1

 OUTPUT count

ENDFOR

END

Question 2:

BEGIN

INPUT year

IF (year MOD 400) = 0 THEN

OUTPUT "Leap year"

ELSE IF (year MOD 4) = 0 AND (year MOD 100) <> 0 THEN

OUTPUT "Leap year"

ELSE

OUTPUT "Not a Leap year"

ENDIF

END

Question 3:

BEGIN

INPUT Number

count = 0

Temp = Number

IF Temp = 0 THEN

count = 1

ELSE

WHILE Temp > 0

Temp = Temp / 10

count = count + 1

ENDWHILE

ENDIF

OUTPUT "Number of digits" = , count

IF count = 1 THEN

OUTPUT "Single-digit number"

ELSE IF count = 2 THEN

OUTPUT "Double-digit number"

ELSE

OUTPUT "Triple-digit or more"

ENDIF

END

Question 4 :

BEGIN

FOR count $\leftarrow$ 1 TO 10

IF (count MOD 3) = 0 THEN

CONTINUE

ENDIF

FOR i $\leftarrow$ 1 TO 10

result = count $\times$ i

OUTPUT count, "x", i, "=", result

ENDFOR

ENDFOR

END

Question 5 :

BEGIN

INPUT number

WHILE number ≥ 10

sum $\leftarrow 0$

Temp = number

WHILE Temp > 0

digit $\leftarrow$ Temp MOD 10

sum $\leftarrow$ sum + digit

Temp $\leftarrow$ Temp / 10

ENDWHILE

OUTPUT "Result = ", sum

number $\leftarrow$ sum

ENDWHILE

OUTPUT "Final single digit = ", number

END